SANJANA RANI

Aerospace Engineer | Aerodynamics, CFD & Experimental Testing | Python

+91-9996673079 | sanjanakamboj98@gmail.com | linkedin.com/in/sanjana-rani | github.com/Sanjanakamboj

PROFESSIONAL SUMMARY
Aerospace engineer with a joint international MSc (France, Belgium, Germany) who works the entire path from concept through to a validated result rather than specializing in one stage of it – between fidelity levels, and between simulation and physical testing on the same problem – always cross-checking a result against an independent method before trusting it. Quick to pick up new tools and domains as a task requires, and enjoys working in small, interdisciplinary teams.

Professional Experience

DLR Institute of Electrified Aero Engines (DLR e.V.), Cottbus, Germany *Feb 2024 - Sep 2024*
Graduate Research Assistant
- Conducted a literature review of electric propulsion systems for UAV/eVTOL applications to select an appropriately rated electric motor for the experimental test setup.
- Collaborated cross-functionally to configure and test a multi-sensor instrumentation array across four spatial positions using DEWESoft DAQ, assessing electric motor-induced EMI effects on microphones and accelerometers for next-generation LU(FT)^2 2030 eVTOL propulsion.
- Generated and processed a multi-microphone acoustic dataset (8 microphones across 6 varied experimental parameters) via a Python pipeline (FFT, coherence, magnitude spectra), analyzing coherence between acoustic and vibration (accelerometer) signals during motor operation – an aeroacoustics/vibroacoustics diagnostic approach – for master's thesis.

Chair of Aerodynamics and Fluid Mechanics, BTU, Cottbus, Germany *Aug 2023 – Feb 2024*
Student Assistant
- Conducted PIV and flow visualisation to characterize flow regimes (laminar, transitional, turbulent) across trial configurations; confirmed laminar flow achievement in final design and produced structured validation reports for department records.
- Designed and iteratively refined a cylindrical plate geometry in Creo – trial-and-error against target laminar-flow geometry constraints, across 3 configurations – then solely assembled the resulting experimental prototype rig (piping, optical/electrical components, laser/camera) for the TEKUS / DEPIK zero-gravity convection project.

Freelance Experience

Freelance CFD Engineer *Mar 2026 – Present*
CFD & Technical Task Development for AI Model Training
- Build and code well-defined technical problems with genuine reference solutions and correctness verifiers for AI model-training platforms, primarily in CFD/OpenFOAM and mechanical engineering domains.
- Deliver against reviewer-graded quality standards across multiple contracted platforms, iterating on structured feedback; numerous tasks accepted to date.

Simulation Lab *May 2023 – Jul 2023*
Aerodynamics Intern (Freelance)
- Ran an 80-case ANSYS Fluent CFD sweep on a provided airfoil geometry (4 angles of attack $\times$ 20 deflection angles), computing lift/drag coefficients (C_L/C_D) to characterize AoA and deflection-angle effects on aerodynamic performance.

Education

Joint MSc in Transfer of Fluid Materials in Aeronautical & Space Applications (TFM-ASA) *Sep 2022 - Sep 2024*
Université de Bordeaux (Bordeaux, France) Université Catholique de Louvain (Louvain-la-Neuve, Belgium) Brandenburgische Technische Universität (Cottbus, Germany)
- **Thesis:** Electromagnetic Interference and its effects on Acoustic Sensor measurements
- **Selected Coursework:** Aerodynamics of External Flows, Gasdynamics & Reacting Flows, Introduction to Turbomachinery, Experiments in Aerodynamics & Fluid Mechanics, Flow Measurements

B.E. in Aerospace Engineering *Aug 2018 - Jun 2022*
Chandigarh University, Gharaun, India
- **Core Coursework:** Aerodynamics I & II, Aerothermodynamics, Propulsion, Wind Tunnel Testing, Flight Mechanics, Aircraft stability & control, Computational Fluid Dynamics
- **Final Project:** CFD Analysis of a hypersonic nozzle using ANSYS Fluent

Key Skills

CFD & Simulation: OpenFOAM, ANSYS Fluent, SU2, XFOIL, GMSH, ParaView, Turbulence Modeling (RANS: k-ω SST, k-ε)

Analytical & Classical Aerodynamics: Thin-Airfoil Theory, Vortex Panel Methods, Prandtl Lifting-Line Theory, Potential Flow

Experimental Test Campaigns: Wind-Tunnel & Water-Channel Testing, Test-Rig Design & Assembly, Instrumentation, DAQ (DEWESoft), PIV
Turbomachinery & Propulsion: Mean-Line Design, Blade Geometry (ParaBlade), Cascade CFD
CAD: SolidWorks, CATIA, Creo
Programming & Software Tools: Python (NumPy, SciPy, Pandas, Matplotlib), MATLAB, Git/GitHub, Docker, LaTeX
Data & Signal Processing: FFT, Coherence Analysis, PSD/Spectrogram Analysis, Digital Filtering, Machine Learning Fundamentals
Other: Orbital Mechanics & Mission Design, AI-Assisted Engineering Workflows (Claude Code)
Languages: English (Fluent), French (Beginner), German (A1, improving)

Selected Projects

UAV Airfoil Selection — **Self-directed** *Aug 2026*

- Selected an airfoil (SD7037) for a small mapping/ISR UAV via XFOIL polar analysis and Prandtl lifting-line finite-wing corrections across 4 candidate airfoils (NACA 2412, Eppler E387, SD7037, S1223), Re = 1.0–3.0$\times 10^5$.
- Found and fixed a real solver-state bug (S1223 convergence collapse) during dataset generation; independently re-verified the finite-wing analysis by hand with 12 unit tests, and flagged the mission's original 9.0 m/s target as sitting at – not above – the estimated stall speed rather than overclaiming margin.

Gas Turbine Design Toolkit — **Self-directed** *Jun – Jul 2026*

- Built a preliminary single-spool turbojet design toolkit mirroring industry chronology: mission profile (cruise Mach, altitude, thrust) $\rightarrow$ 0D/1D Brayton-cycle sizing $\rightarrow$ compressor/turbine mean-line design $\rightarrow$ 3D blade geometry (free-vortex twist, ParaBlade) $\rightarrow$ cascade CFD (GMSH+SU2/OpenFOAM) $\rightarrow$ off-design performance across the flight envelope.
- Backed by 503 automated tests grounded in hand-derived analytical relations or real solver output, not mocks; explicitly scoped out mechanical/structural design and unverified emissions prediction rather than overclaiming coverage.

Water-Channel PIV Study — Airfoil & Cylinder — BTU Cottbus *Aug 2023 – Feb 2024*

- Conducted water-channel flow visualization and stereoPIV (PIVlab) on airfoil and cylinder models, using tracer particles and laser-sheet illumination to extract quantitative streamline and velocity fields.
- Used the water channel's variable flow speed to compute C_l, C_d, and C_p across both geometries.

Flow Past a Cylinder — From-First-Principles CFD Study — **Self-directed** *Aug 2025 – Feb 2026*

- Built the full research pipeline from physics through validation: analytical Reynolds-regime derivation, a from-scratch Python Navier-Stokes solver, then an OpenFOAM study (Re = 10–1000) with a 3-level mesh-independence check.
- Automated case generation and force/FFT extraction (Python) across the Reynolds sweep, building the pipeline to benchmark Strouhal number and drag coefficient against published literature.

Extracurricular Activities

Aerodynamics Club, Chandigarh University – *2018 – 2022* – team-built RC aircraft and static wind-tunnel models as part of coursework projects
Bhangra Club, Chandigarh University – *2018 – 2022* – competed at zonal-level dance competitions
Cottbus Beats (Dance), BTU Cottbus – *Aug 2023 – Sep 2024* – performed at community events, including volunteering at a local old-age home and the Cottbus summer festival

FlowSleuth — Agentic CFD Root-Cause Diagnostics — **micro1 Frontier Engineering Challenge 2026 (Hackathon)** *Aug 202*

- Built an agentic diagnostic tool for failed/stalled OpenFOAM CFD runs during a 3-day hackathon: returns a ranked root cause with exact file/line evidence, a bounded remediation plan, and calibrated uncertainty, gated behind an explicit human-approval step before any changes.
- Iteratively validated the diagnostic agent against a 12-case synthetic OpenFOAM benchmark (causal ordering, evidence specificity, confidence calibration), improving from a 63.42/100 deterministic-keyword baseline to 98.48/100 (+35.06 pts); an independent locked external CFD review confirmed 12/12 root-cause agreement on sealed labels.